
YUE YU

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RESEARCH INTERESTS

My research focuses on building reliable and efficient large language models (LLMs) with reasoning capabilities. Specifically, I am interested in:

- **Post-training and Alignment:** Reinforcement learning, preference optimization, and reward modeling for improving alignment, instruction following, and non-verifiable tasks.
- **Retrieval- and Tool-Augmented Learning:** Designing frameworks that integrate external knowledge and tools into LLM reasoning, including retrieval-augmented generation (RAG), tool use, and judge models.
- **Complex Reasoning and Coding:** Advancing the capability of LLMs in coding-related tasks, including competitive programming, code editing/debugging, and test generation.

PROFESSIONAL EXPERIENCE

CodeGen Team, Meta Fundamental AI Research (FAIR)

AI Research Scientist, Managers: Taco Cohen, Gabriel Synnaeve
Working on Meta's Next Generation LLMs (Avocado).

Menlo Park, CA, USA
(FTE) Dec 2025 -

AI Frontier Reasoning Team, Meta Superintelligence Labs (Previous Meta GenAI)

AI Research Scientist, Managers: Melanie Kambadur, Rui Hou, Maryam Fazel-Zarandi

Menlo Park, CA, USA
(FTE) Jan 2025 -

I am a core contributor of Llama 4 model with a focus on coding, reasoning, and instruction following capabilities.

Projects I: Post-training for Coding: Developed a post-training pipeline (SFT+RL) for improving competitive programming and generalized coding. Improved LiveCodeBench by 45%, Aider Polygrot by 33%; Code Editing by 27%.

Projects II: Code Test Generation: Developed recipes to enhance LLM performance on code I/O prediction and unit test generation for competitive programming. Improved CruxEval accuracy by 26%, and Internal TestGen by 74%.

Projects III: Instruction Following SFT Data Generation with Thinking Traces: Designed a data creation pipeline including *difficulty-aware prompt filtering, thinking trace and response generation, and judge-based response filtering*. Improved IFEval by 8% and internal non-verifiable IF by 5.3%.

Projects IV: RL for Instruction Following with Rubrics-Based Judge: Developed *synthetic rubrics* and applied rubrics-based judges to enhance *non-verifiable* instruction following through reinforcement learning. Improved IFEval by 8.2%, IFBench by 5.2% and internal non-verifiable IF by 11.1%.

Llama Team, Meta GenAI

Research Intern, Host: Rui Hou, Manager: Melanie Kambadur

Menlo Park, CA, USA
(Intern) May - Dec. 2024

Topic: Generative Reward Models (With one paper in [NAACL 2025](#)).

Applied Deep Learning Research Team, NVIDIA

Research Intern, Host: Wei Ping, Manager: Mohammad Shoeybi

Santa Clara, CA, USA (Remote)
Jan. 2024 - May 2024

Topic: Building Instruction Finetuning Recipe for Improving RAG Capability of Large Language Models (With one paper in [NeurIPS 2024](#)).

News Understanding Team, Google Research (Now Google DeepMind)

Research Intern, Host: Jiaming Shen, Co-host: Tianqi Liu, Zhen Qin, Manager: Jialu Liu

New York City, NY, USA
May 2023 - Aug. 2023

Topic: Empowering Large Language Model In-context Learning with Free-text Rationales (With one paper in [ACL 2024](#)).

Productivity and Intelligence Group, Microsoft Research

Research Intern, Mentor: Chenyan Xiong, Manager: Arnold Overwijk

Redmond, WA, USA
May 2021 - Aug. 2021

Topic: Zero-shot Learning for Generalizable Dense Text Retrieval. (With one paper in [EMNLP 2022](#)).

EDUCATION

School of Computational Science and Engineering, Georgia Institute of Technology
Ph.D. in Computational Science and Engineering

Atlanta, GA, USA
Aug. 2019 - Dec. 2024

• Ph.D. Advisor: [Dr. Chao Zhang](#);

• **Research Interest:** Large Language Models, (Tool/Retrieval) Augmented Learning, Reasoning, Data-centric AI.

- **Core Courses:** Computational Data Analysis, Mathematical Foundation of Machine Learning, Algorithms, Convex Optimization, Modeling and Simulation, Deep Learning for Text Data.
- **Thesis Topic:** Towards Efficiently and Effectively Harnessing Large Pre-trained Models via Data-centric Lens.

Department of Electronic Engineering, Tsinghua University
B.Eng. in Electronic Engineering

Beijing, China
Aug. 2015 - July 2019

- Research Assistant in the Future Internet & Communication Lab advised by [Dr. Yong Li](#);

SELECTED PUBLICATIONS

(I have published in premium ML and NLP conferences with more than 3000 citations. The full publication list can be found in [this link](#), * stands for equal contribution):

1. **Yue Yu**, Zhengxing Chen, Aston Zhang, Liang Tan, Chenguang Zhu, Richard Yuanzhe Pang, Yundi Qian, Xuewei Wang, Suchin Gururangan, Chao Zhang, Melanie Kambadur, Dhruv Mahajan, and Rui Hou. “[Self-Generated Critiques Boost Reward Modeling for Language Models](#)”. In *Proceedings of the 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
2. **Yue Yu**, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi, and Bryan Catanzaro. “[RankRAG: Unifying Context Ranking with Retrieval-Augmented Generation in LLMs](#)”. In *Proceedings of the Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
3. **Yue Yu**, Jiaming Shen, Tianqi Liu, Zhen Qin, Jing Nathan Yan, Jialu Liu, Chao Zhang, Michael Bendersky. “[Explanation-aware Soft Ensemble Empowers Large Language Model In-context Learning](#)”. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
4. **Yue Yu***, Yuchen Zhuang*, Jieyu Zhang*, Yu Meng, Alexander Ratner, Ranjay Krishna, Jiaming Shen, Chao Zhang. “[Large Language Model as Attributed Training Data Generator: A Tale of Diversity and Bias](#)”. In *Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks (NeurIPS)*, 2023.
5. Yuchen Zhuang*, **Yue Yu***, Kuan Wang*, Haotian Sun, Chao Zhang. “[ToolQA: A Dataset for LLM Question Answering with External Tools](#)”. In *Proceedings of the Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks (NeurIPS)*, 2023.
6. **Yue Yu**, Rongzhi Zhang, Ran Xu, Jieyu Zhang, Jiaming Shen and Chao Zhang. “[Cold-Start Data Selection for Few-shot Language Model Fine-tuning: A Prompt-Based Uncertainty Propagation Approach](#).” In *Proceedings of the 61th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
7. **Yue Yu**, Chenyan Xiong, Si Sun, Chao Zhang, and Arnold Overwijk. “[COCO-DR: Combating the Distribution Shift in Zero-Shot Dense Retrieval with Contrastive and Distributionally Robust Learning](#).” In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
8. **Yue Yu**, Lingkai Kong, Jieyu Zhang, Rongzhi Zhang, and Chao Zhang. “[AcTune: Uncertainty-Based Active Self-Training for Active Fine-Tuning of Pretrained Language Models](#).” In *Proceedings of the 2022 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2022.
9. **Yue Yu***, Simiao Zuo*, Haoming Jiang, Wendi Ren, Tuo Zhao and Chao Zhang, “[Fine-Tuning Pre-trained Language Model with Weak Supervision: A Contrastive-Regularized Self-Training Approach](#)”, In *Proceedings of the 2021 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2021.

HONORS AND AWARDS

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| • NeurIPS 2023 Scholar Award | Nov. 2023 |
| • Best Paper Award at Machine Learning for Health 2022 | Nov. 2022 |
| • ACM SIGKDD Student Registration Award | Aug. 2020 |
| • Excellent Graduate, Tsinghua University & Beijing City (Top 2% over 3292 graduate students) | July 2019 |
| • Comprehensive Scholarship, Tsinghua University (Top 1%) | Oct. 2018 |
| • Award from Tsinghua University Initiative Scientific Research Program (5000 USD) | May 2018 |
| • Comprehensive Scholarship, Tsinghua University (Top 5%) | Oct. 2016, Oct. 2017 |

PROFESSIONAL SKILLS

- Programming language: C++ , Python, MATLAB, Latex.
- Deep learning frameworks: Keras, Pytorch.

SERVICES

- **Area Chair:** NeurIPS 2025, ACL 2024/2025, NAACL 2025, EMNLP 2024.
- **Conference Program Committee:** NeurIPS 2022/2023/2024; ICML 2022/2024/2025; ICLR 2024/2025; ACL 2023; KDD 2023; IJCAI 2023; EMNLP 2022/2023.